A Treasure Map Points to Missing Fossils

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Fossils show us life from millions of years ago. But turning into a fossil is very rare, so only a tiny number of creatures have been saved. Scientists can only find one where the rock is open to view.

That leaves big gaps in what we know about life long ago. Some times and places have given us lots of **fossils**, others almost none. Digging takes time and money, so knowing where to look helps.

Two scientists have made a tool to help. Thomas A. Neubauer and Mathias Harzhauser studied **shallow-sea fossils** from the past 540 million years, and mapped where fossil clues are missing.

They used the **Paleobiology Database**, a huge free list of fossil finds. They also looked at rocks that could hold more **fossils**.

The results were a surprise. The oldest times have already been studied a lot, and the best chances for new finds are in more recent times: the **Cenozoic Era**, which began 66 million years ago.

Scientists marked spots where unknown **fossils** are most likely. Their map helps **fossil hunters** plan better digs.

Word Treasure

fossils -- The saved remains of animals and plants from long ago.

Cenozoic Era -- The time from 66 million years ago until now.

Paleobiology Database -- A huge free list of fossil finds.

shallow-sea -- In the shallow water of a sea.

fossil hunters -- People who search for fossils.

Background you need

Fossils are the remains of animals and plants that lived long ago, saved in rock.

Only a tiny number of creatures ever become fossils.

Scientists study fossils to learn what life was like on Earth millions of years ago.

Break it down

WHO: Two scientists, Thomas A. Neubauer and Mathias Harzhauser
WHAT: A map that points to where missing fossils may be found
WHERE: In shallow seas around the world
WHY: Digging takes time and money, so knowing where to look helps

Why it matters

Fossils are like a time machine that shows you what animals looked like millions of years before you were born, and new finds could add missing pages to that story.

Test what you remember

1. Why are fossils rare?

- (A) Turning into a fossil is very rare
- (B) Rocks are too hard
- (C) Animals never die
- (D) Scientists hide them

2. Who made the new tool?

- (A) Thomas A. Neubauer and Mathias Harzhauser
- (B) Mary Anning
- (C) Albert Einstein
- (D) Charles Darwin

3. What did they study?

- (A) Rocks on the moon
- (B) Dinosaur bones in museums
- (C) Fish in aquariums
- (D) Shallow-sea fossils from the past 540 million years

4. What list did they use?

- (A) The phone book
- (B) A school library
- (C) A weather map
- (D) The Paleobiology Database

5. Where are the best chances for new finds?

- (A) In the desert
- (B) In more recent times, the Cenozoic Era
- (C) In deep space
- (D) In the oldest times

6. What does the map help fossil hunters do?

- (A) Write books
- (B) Build houses
- (C) Fly planes
- (D) Plan better digs

Answer key (try the quiz first!): 1: A 2: A 3: D 4: D 5: B 6: D

Questions to think about

MY THOUGHTS (at least 20 words):